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Baseline metagenome-assembled genome (MAG) data of Sikkim hot springs from Indian Himalayan geothermal belt (IHGB) showcasing its potential CAZymes, and sulfur-nitrogen metabolic activity

Sayak Das^{1,2} · Ishfaq Nabi Najar¹ · Mingma Thundu Sherpa¹ · Santosh Kumar¹ · Prayatna Sharma¹ · Krishnendu Mondal³ · Sonia Tamang¹ · Nagendra Thakur¹

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Abstract

Here we present the construction and characterization of metagenome assembled genomes (MAGs) from two hot springs residing in the vicinity of Indian Himalayan Geothermal Belt (IHGB). A total of 78 and 7 taxonomic bins were obtained for Old Yume Samdong (OYS) and New Yume Samdong (NYS) hot springs respectively. After passing all the criteria only 21 and 4 MAGs were further studied based on the successful prediction of their 16 S rRNA. Various databases were used such as GTDB, Kaiju, EzTaxon, BLAST XY Plot and NCBI BLAST to get the taxonomic classification of various 16 S rRNA predicted MAGs. The bacterial genomes found were from both thermophilic and mesophilic bacteria among which Proteobacteria, Chloroflexi, Bacteroidetes and Firmicutes were the abundant phyla. However, in case of OYS, two genomes belonged to archaeal *Methanobacterium* and *Methanocaldococcus*. Functional characterization revealed the richness of CAZymes such as Glycosyl Transferase (GT) (56.7%), Glycoside Hydrolase (GH) (37.4%), Carbohydrate Esterase family (CE) (8.2%), and Polysaccharide Lyase (PL) (1.9%). There were negligible antibiotic resistance genes in the MAGs however, a significant heavy metal tolerance gene was found in the MAGs. Thus, it may be assumed that there is no coexistence of antibiotic and heavy metal resistance genes in these hot spring microbiomes. Since the selected hot springs possess good sulfur content thus, we also checked the presence of genes for sulfur and nitrogen metabolism. It was found that MAGs from both the hot springs possess significant number of genes related to sulfur and nitrogen metabolism.

Keywords Metagenome-assembled genome · Sikkim · Data binning · CAZymes · Uncultured microbiota

Introduction

The microbial communities residing in various spheres of the Earth are continually deciphered by sophisticated deep throughput sequencing techniques vis-a-vis microbiome analysis (Deb and Das 2022; Bennett et al. 2022; Zhong et al. 2022). Their diversification and ecological role are magnanimous and revealing their taxonomy is itself an intrigue concern for biologists even today. Prokaryotes are diverse and known to dominate almost all the spheres of life (Whitman et al. 1998). They play crucial roles in biogeochemical processes that is a pre-requisite for life (Falkowski et al. 2008). They also can reside in extreme environments such as thermophiles living at hot temperatures, psychrophiles living in cold habitats, halophiles living in high saline regions or barophiles dwelling in the varying depths of oceans. One of the important extreme environmental habitats are the

Sayak Das and Ishfaq Nabi Najar contributed equally to this work.

✉ Nagendra Thakur
nthakur@cus.ac.in

¹ Department of Microbiology, School of Life Sciences, Sikkim University, 6th Mile, Samdur, Gangtok, Sikkim 737102, India

² Department of Life Science and Bioinformatics, Hargobind Khurana School of Life Sciences, Assam University, Silchar, Assam 788011, India

³ Department of Microbiology, Vidyasagar University, Midnapore, West Bengal 721102, India

geothermal fields on the surface of earth and within hydrosphere. Geothermal fields possess many features such as hot springs, fumaroles, geysers and mud pits. Hot springs are the widely studied features worldwide with respect to geochemistry and microbiology. These are the geothermal points on the surface of earth through which hot water emerges out having temperature 10 degrees higher than the atmospheric temperature. Hot springs are dominated by mainly thermophilic microorganisms from both the bacterial and archaeal domains. Conventional methodology has been extensively used to explore the microbial diversity in such extreme habitats. However, by utilizing conventional methodology only a small fraction of the bacterial diversity (<1%) can be isolated and cultivated (Amann et al. 1995). As a result, non-cultivation methodologies for extracting genomes from any resources have arisen as valuable complements to established microbiological methods. Techniques like meta-omics has evolved as a key tool to identify microorganisms, gene functions, and metabolic pathways. Shotgun metagenomics is a technology that allows scientists to quickly examine microbial populations by sequencing a random fraction of genetic material from the environment (Papudeshi et al. 2017). Besides metagenomics which gives the framework of random taxonomic impressions within total environmental DNA, many biogeochemical, ecological, and evolutionary studies necessitate data organization into genomes (Alneberg et al. 2018). This is important because, while individual genes or genome fragments can provide helpful information about a community's metabolic capability, most biochemical changes occur inside a cell, requiring set of genes organized in controlled pathways (Bult et al. 1996; Wu et al. 2009; Sunagawa et al. 2015). This can be overcome by the process of binning. Binning allows contigs or scaffolds from the same lineage to be clustered and genomes to be rebuilt (Abe et al. 2005; Spang et al. 2015). By assembling short shotgun reads into longer contigs, which are then sorted into groups, or bins, of contigs originated from the same organism, genomes can be retrieved from metagenomes. These are then submitted into many worldwide public repositories which are showing growth at double the rate per year in the Genomes On Line Database (GOLD) (Reddy et al. 2015). The constructed bins are referred to as metagenome-assembled genomes (MAGs) (Hugerth et al. 2015).

This huge volume of sequence data can be used to predict worldwide phylogenetic lineage distributions, metabolic potentials and functional attributes of different microbial communities. This enables estimating the possible contributions of populations to the community processes. Various studies have been done so far on many environmental or biological samples such as alkaline sulphidic tailings (Li and Li 2021), the Baltic Sea (Hugerth et al. 2015), the gut

microbiota of Tibetan pig (Zhou et al. 2020) etc. MAGs are now used as the most important prediction tool to harness the unknown microbial biodiversity, their enzymatic potential, and ecological community profiling (Jiménez-Volkerink et al. 2023; Lynes et al. 2023; Rao et al. 2023; Qu et al. 2023; Zhou et al. 2022; Allieux et al. 2022; Chen et al. 2022; Yin et al. 2022). Many studies have shown that the MAGs were taxonomically assigned to various bacterial phyla such as Proteobacteria, Bacteroidetes, Firmicutes, Spirochetes, Verrucomicrobia, Nitrospirae and Fibrobacterota and many other archaeal phyla (Zhou et al. 2020). Further, the functional characterization of various carbohydrate active enzymes (CAZymes), metabolic pathways, biochemical characterization such as heavy metal analysis, and antimicrobial profile have been deduced from MAGs (Alneberg et al. 2014; Li and Li 2021; Zhou et al. 2020). Other novel metabolic pathways and taxons have also been successfully reconstructed by MAGs study (Rožanov et al. 2021; Speth et al. 2022; Kim et al. 2022; Pan et al. 2022).

Besides many studies done, however, geothermal features have been less studied. In the present study, we have used high throughput sequencing technology to generate MAGs and have evaluated many functional and metabolic pathways such as CAZymes, sulfur and nitrogen metabolism, heavy metals and antibiotic profile of various MAGs generated. CAZymes are very important as they are the main degraders of polysaccharides, glycoconjugates, and indigestible oligosaccharides to fermentable monosaccharides (Kaoutari et al. 2013). The major carbohydrate active enzymes able to degrade complex carbohydrates are glycoside hydrolases, polysaccharide lyases, and carbohydrate esterases (Abot et al. 2016; Kaoutari et al. 2013). Glycoside hydrolases and polysaccharide lyases can cleave glycosidic bonds in cellulose, xylan, pectin, hyaluronan, and heparin, among other carbohydrate substrates (Kaoutari et al. 2013). Thus, these enzymes are very important with respect to industrial settings and are of higher significance to study. Similarly, study of antibiotic resistance is very important as antibiotic-resistant superbugs are posing a global danger to morbidity and mortality of human life (Tacconelli et al. 2018). However, antibiotic resistance profile of various geothermal features such as hot springs are very less studied ((Najar et al. 2020b). The present study is important as it highlights the antibiotic resistance profile of hot spring habitats. Also, it shows the influence and abundance of heavy metal tolerant genes and their co-occurrence with the antibiotic counterparts.

In the present study, two hot springs were considered for investigation based on their high temperature and altitude. These hot springs are located in North Sikkim Himalayan region. These are more or less pristine with lesser anthropogenic activities present. These hot springs are solfataric

in nature with considerably higher sulfur content which can be also predicted with their sulfur odor. Previous studies on NYS and OYS solfataric sediments explored the other complex biomes present in the hot spring ecosystem such as viruses (Das et al. 2020a), archaea (Das et al. 2020b) and fungi (Das et al. 2021). Keeping this into consideration, sulfur and nitrogen metabolism was studied in the generated MAGs from these hot springs. Thus, the main aim of the study was to create MAGs to check the phylogenetic relationship and to functionally characterize the genomes. According to the literature review, this is the first ever research on MAGs derived from metagenomic samples conducted on Indian hot springs.

Materials and methods

Sampling site and collection

Soil samples were collected from the two physically closely located hot springs of Sikkim based at 4500mamsl (meters above the mean sea level) at Yume Samdong valley. The hot

springs under study were – Old Yume Samdong (OYS) and New Yume Samdong (NYS) hot springs. The geographical position of the coordinates and the elevation of the hot springs were measured with the help of GPSMAP 78 S (Garmin, India) and mapped through ARC-GIS Google Earth (Fig. 1). The *in-situ* hot spring conditions were recorded by HORIBA, Japan – an automated portable field instrument for analyzing water parameters. The solfataric mud samples (1000gm) were aseptically collected (Wang et al. 2014) from the hot springs of OYS and NYS in sterile sample containers. The samples were pooled from ten different spots encompassing the hot springs. From each spot of the sampling site, approximately 100gm of mud-soil was excavated and kept in the sampling container, where the final weight obtained was 1000gm per hot spring. In order to have a complete representation of the hot spring, the samples were pooled from all the perimeters of the hot springs. They were collected in triplicates. Thus, overall, from both the sampling sites, 6000gm (6000gm = 1000gm × 3 triplicates × 2 hot springs) of solfataric mud samples was collected across 20 different sampling sites. The samples were preserved *in-situ* by storing in the thermally insulated sampling box

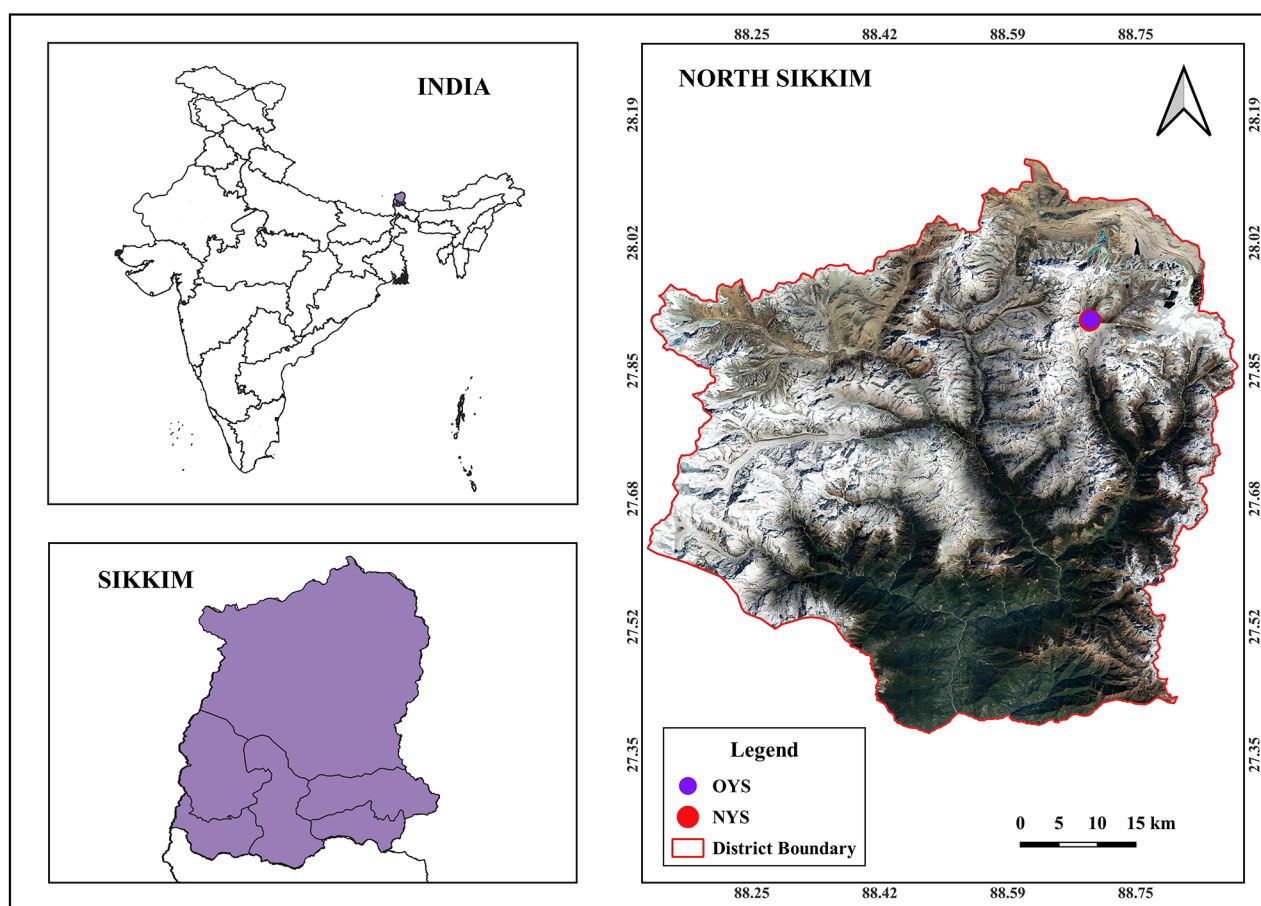


Fig. 1 Map showing the location of sampling sites, OYS (Old Yume Samdong) and NYS (New Yume Samdong).

packed with dry ice and ice gel bags. Later, they were carefully transported (temperature being maintained at 4 °C) to the laboratory for DNA extraction analysis.

Shotgun metagenomic sequencing

Total environmental DNA (eDNA) was extracted from both the soil samples by using Illumina TruSeq® DNA Library Prep Kits (Wang et al. 2014). The quality of eDNA of both the samples were analyzed using Agilent 2100 Bioanalyzer and their quantity was checked by using Nanodrop. NYS and OYS eDNA were subjected to standard paired end DNASeq library preparation using Illumina 4000 as per the manufacturer's protocol. Samples were sequenced using 101 bp paired end module sequencing. After sequencing, reads obtained for each library (NYS and OYS), were then subjected to quality control using NGSQC tool kit (Patel and Jain 2012) to get the high quality (HQ) reads in fastq format from the two samples. Stringent criteria of 80:30 was used to obtain high quality filtered reads wherein more than 80% HQ bases each having phred scores ≥ 30 was considered for building genome assembly. HQ filtered paired end libraries of the samples of NYS and OYS were independently used to generate respective genome assembly of the samples using a de Bruijn graph-based assembler MetaSPAdes (Nurk et al. 2017) using default parameters.

Metagenomic assembly and binning

An integrative workflow was used to reconstruct microbial genomes from metagenomic data. Individual sample assembly, binning, de-replication and removal of contig contamination was done as per the methods described previously (Olm et al. 2017; Stewart et al. 2019; Uritskiy and Di Ruggiero, 2019). Metagenomic reads per sample were corrected by BayesHammer and then assembled using MetaSPAdes v3.13.0 (Nurk et al. 2017). The *de novo* assembly was performed using default parameters. Thus, the assembled data was segregated into different microbial Genome obtained from MetaSPAdes for the samples respectively. To segregate the different microbial genomes from the assembled NYS and OYS metagenome assembly, MAXBIN 2.0 (Wu et al. 2016) was used using default parameters. Genome binning was carried out for all the contigs longer than 2 kb in size. Further refinement on both sets of bins were performed to produce a combined and robust bin set using the MetaWRAP (Uritskiy and Di Ruggiero, 2019) Bin refinement module with the parameters -m 150 -t 18 -c 50 -x 10. All the resulting genome bins ($n=1,095$) from the samples were then pooled together and de-replicated using the dRep package v2.3.2 with the following parameters: -comp 80 -con 10 -strW 1 -p 36 -S_algorithm ANImf -length 100,000

-checkM_method lineage_wf (Olm et al. 2017). The putative contig contamination per genome was detected and removed by using MAGpurify v2.1.2 (Nayfach et al. 2019). Genome quality was estimated using CheckM v1.0.12 to calculate genome completeness and contamination (Parks et al. 2015). On the basis of completeness, contamination of individual bin, bin with 80% completeness and 10% contamination was considered as putative individual genome and was screened for presence of 16S rRNA gene for binned genome. All the bins were checked for the number of marker genes until no bins could split further. After the EM process, MAXBIN scanned the 107 marker genes (these genes are known to be present as single copy gene in bacteria) in all bins for genome completeness, which is measured as the fraction of unique marker genes versus all marker genes.

Taxonomic and phylogenetic analysis of MAGs

Taxonomic lineages of the putative MAGs were inferred using the recently developed database GTDB release v95 (Parks et al. 2018) and the relevant toolkit GTDB-Tk v1.2.0 (Chaumeil et al. 2019). Based on domain-specific reference trees, GTDB implemented a new microbial taxonomy which can be used for robust classification of the query genomes by GTDB-tk. Taxonomy was also assigned to the MAGs for each sample using Kaiju v. 1.6.267 with the NCBI BLAST non-redundant protein database *nr* v. 2017-05-16 (Menzel et al. 2016). Kaiju was run using Maximum Exact Matches (MEM) mode with eleven substitutions allowed with an e-value cut-off for taxonomic assignment of 0.01. to the MAGs with ambiguity in taxonomic classification, 16S rRNA sequences were extracted using RAST platform (Aziz et al. 2008) and then BLAST was performed to the bins against EzTaxon (Yoon et al. 2017), NCBI BLASTn, BLAST XY Plot databases (Pedraza-pérez et al. 2018). All the above four methods were used to predict the exact taxonomic lineages to all the bins screened. For phylogenetic analysis the related genomes were extracted from the NCBI data base. Then the MAGs and their related genome sequences were aligned by Clustal Omega (Thompson et al. 1994). Maximum likelihood phylogenetic tree was constructed using MEGA 11 software with bootstrap values of 500. The phylogenetic tree was the edited and modified in iTOL interactive tree of life (Letunic et al. 2021).

Gene prediction of binned genome

The final filtered true binned genome from both the samples (NYS and OYS) were then used for gene prediction, annotation and downstream screening of these genomes using RAST SEED viewer (Aziz et al. 2008). The gene prediction for protein coding genes (ORFs) was predicted using

Prodigal (Hyatt et al. 2010). ORFs shorter than 180 nucleotides were excluded from further analysis by default. tRNA genes were predicted with the ARAGORN program (Laslett and Canback 2004) and rRNA genes (5S, 5.8S, 16S, 18S, 23S, 28S) were identified and classified using RNAmmer (Lagesen et al. 2007). All ORFs were searched against KEGG Orthology (KO) terms (Kanehisa et al. 2012) and Gene Ontology (GO) terms to predict the gene functional classification (Tatusov et al. 2003). ORFs without any search outcomes were annotated as “hypothetical protein”.

Functional characterization of ARGs, MRGs and CAZymes

The predicted amino acid sequence from binned genome for both the NYS and OYS samples were used to check the presence of antibiotic resistance gene using CARD database (Alcock et al. 2020). Heavy metal gene screening of binned genome of NYS and OYS was done using the BacMet database (Pal et al. 2014). For each genome, the genes encoding CAZymes were predicted and annotated using dbCAN2 (Zhang et al. 2018), which curated a set of HMMs corresponding to the CAZyme families created by the CAZy database (Lombard et al. 2014). HMMER v3.1b2 (Finn et al. 2011) was used to search the HMMs of dbCAN2, and the result was filtered with the thresholds below: E-value < 1e-15 and coverage > 0.35. An independent pipeline using RefSeq database available in Kbase was used to extract various CAZymes from Metagenome-Assembled Genomes (MAGs) (Arkin et al. 2018).

Assessment of sulfur and nitrogen metabolism

MAGs were analyzed for sulfur and nitrogen metabolism. The whole genome sequences were uploaded in SEED RAST server and functional analysis was done. The abundance of various genes responsible for sulfur and nitrogen metabolism were represented in heat maps.

Data availability and accession

Raw metagenomic reads were submitted to Sequence Read Archive (SRA), NCBI to obtain the Bio-Sample and Sequence Read Archive (SRA) accession numbers. The OYS data, Bio-Project accession obtained is PRJNA485728; Bio-Sample accession is SAMN09813281 and SRA is

SRP158027 for the sample name OYSMUD4. The NYS data Bio-Project accession obtained is PRJNA485701; Bio-Sample accession is SAMN09813022 and SRA is SRP158029 for the sample name NYSMUD4.

Results

Study area

The hot springs located at the North Sikkim district of Sikkim were selected for the current study. Sikkim is a landlocked mountainous state of India constituting the North-east region. This field study is located at Lachung at the altitude above 4500 mamsl where the average annual temperature is around 1.7 °C and the rainfall is approximately 305 mm. Basically this region has Tundra climate and experiences heavy snowfall from September to June. It maintains a cold snowy and windy atmosphere throughout the year. Being located at such high altitude it also experiences uv radiation having uv index 2. It is also the basecamp territory for many glaciers such as Changme Khang, and Chagme Khangpu (Sherpa et al. 2020). Both the NYS and OYS hot springs are closely located in the same geographical vicinity with the latter being near the river Lachung chu Fig. 1. They are pristine and have bubbles coming beneath their soil fissures. The temperature of the NYS hot spring was around 57 °C and in case of OYS hot spring was 61 °C during sample collection. Both of them were rich in sulfurous odor. The *in-situ* parameters measured were similar to both the hot springs and this may be believed that they share a common point of geothermic origin (Table 1). Both the hot springs were alkaline and sulfurous in nature. The mud extracted from both the hot springs had similar geophysical characteristics and approximately shared the same geochemistry (Das et al. 2020c). Thus, it is very fascinating phenomena where the hot springs showing average temperature of 59 °C and the natural atmospheric temperature hovering at around – 2 to 3 °C at the time of sample collection.

Quality filtering and assembly

A total 49,570,898 (49.57 million) paired end reads for NYS sample and 56,681,402 (56.68 million) paired end reads for OYS sample was generated using the Illumina sequencing method. Out of 49,570,898 generated sequencing reads

Table 1 On-site physicochemical properties of NYS-OYS solfataric mud ecosystem

Hot Springs	Latitude	Longitude	Elevation	Temperature	pH	Fluoride (as F in gL ⁻¹)	Chloride (as Cl in gL ⁻¹)	Sulphite (as Na ₂ SO ₃ in gL ⁻¹)
NYS	27.917302°N	88.694308°E	4665 m	57 °C	9.1	0.1	0.5	0.7
OYS	27.918242°N	88°694,935°E	4668 m	61 °C	8.77	0.2	1.0	0.8

Measured on site through HiMedia Aqua Check Test Kit, India (colorimetric test)

for NYS sample, 34,594,204 (34.59 million) HQ reads with phred score ≥ 30 was obtained after quality filtering. Similarly, out of 56,681,402 generated sequencing reads for OYS, 40,319,936 (40.32 million) HQ reads with phred score ≥ 30 was obtained after quality filtering. These HQ reads were processed further for taxonomy classification and abundance analysis by using various databases such as GTDB and Kaiju for understanding the microbiome data. The NYS and OYS hot spring solfataric mud sediments, had all the three domains of life and viruses in it. In the NYS hot spring, Archaea had 3789 reads ($\sim 0.03\%$), Eukarya had 4,956,671 reads ($\sim 39.8\%$), Bacteria had 7,528,447 reads ($\sim 60.2\%$) and virus had 16,958 reads ($\sim 0.17\%$) whereas from the OYS hot spring mud sediments, Archaea had 56,944 reads ($\sim 1.06\%$), Eukarya had 24,508 reads ($\sim 0.5\%$), Bacteria had 5,296,073 reads ($\sim 98.42\%$) and virus had 3596 reads ($\sim 0.07\%$) respectively. A single raw genome fasta file of 67,950,615 bp (~ 67 Mbp) assembly size and with 38,986 scaffolds was assembled for NYS sample. Similarly, single raw genome fasta of 428,411,900 bp (~ 428 Mbp) assembly size with 456,587 scaffolds was assembled for OYS sample. The G + C content of the NYS and OYS samples were 47.73% and 55.23% respectively as shown in (Table 2).

Segregation of assembled data

Out of total 38,986 metagenomic scaffolds for NYS and 456,587 metagenomic scaffolds for OYS sample, 7 different taxonomic bins for NYS and 78 different taxonomic bins for OYS were obtained. On the basis of completeness, contamination of individual bin, bin with 80% completeness and 20% contamination was considered as putative individual genome and was screened for the presence of 16 S rRNA gene for binned genome.

From seven binned genome of NYS sample, five bins passed the criteria filter of 80% completeness and 20% contamination. These five binned genomes were used to predict the 16S rRNA gene using tool RNAmmer v1.2 (Lagesen et al. 2007) and RAST Platform (Aziz et al. 2008). Out of five binned genomes only in four bins, 16S rRNA gene was successfully predicted. These four represented the true genome present in NYS sample Supplementary Table 1. Similarly,

for OYS sample out of 78 bins, only 34 bins passed the criteria filter of 80% completeness and 20% contamination. Out of 34 bins, 16S rRNA gene was successfully predicted in 21 bins. These 21 bins represented the true genome present in OYS sample Supplementary Table 2.

Predicted gene abundance in binned genomes

The average number of scaffolds among various genomic assemblies were 220.4 scaffolds. The N50 value vary for all the different assemblies. The average protein predicted for various assemblies was found to be 3469.4. The number of non-coding proteins were less as the average non-coding proteins among assemblies were 48 as compared to coding proteins as shown in Supplementary Table 3.

Taxonomic identification of binned genomes

The predicted 16S rRNA gene sequence from all the binned genomes for NYS and OYS was used to find the nearest references for the binned genome using various databases such as GTDB, Kaiju, EzTaxon (Yoon et al. 2017) and NCBI BLAST non-redundant protein database *nr*. It was found that almost all the databases showed similar results and the common taxonomic identities among at least three databases were taken into consideration. The results have shown that in case of OYS, MAGs comprise of both the domains such as Archaea and Bacteria. However, in NYS, MAGs from only bacterial domain represent the true genomes. The phylum Proteobacteria and Firmicutes were equally predominant in NYS whereas, Chloroflexi followed by Proteobacteria, Bacteroidetes and Planctomycetes were predominant in case of OYS. In case of OYS, two genomes corresponded to the phylum Euryarchaeota of archaeal domain comprising of genera *Methanobacterium* and *Methanocaldococcus*. Besides this there was considerable distinction at different hierarchical levels in all the MAGs such as at order, class, family, genus, and species as shown in Table 3. The species related to phylum Proteobacteria include *Desulfobacca acetoxidans*, *Desulfonema magnum*, *Syntrophobacter fumaroxidans*, *Serratia surfactantifaciens*, and *Acinetobacter baumannii*. Moreover, the species related to phylum Chloroflexi include *Chloroflexus aurantiacus*, *Caldilinea aerophila*, *Bellilinea caldifistulae*, and *Thermomarinilinea lacunofontalis*. Maximum likelihood phylogenetic tree was constructed using MEGA 11 software with bootstrap values of 500. The phylogenetic tree was edited and modified in iTOL interactive tree of life as shown in Fig. 2.

Table 2 Summary of Primary Genome Analysis

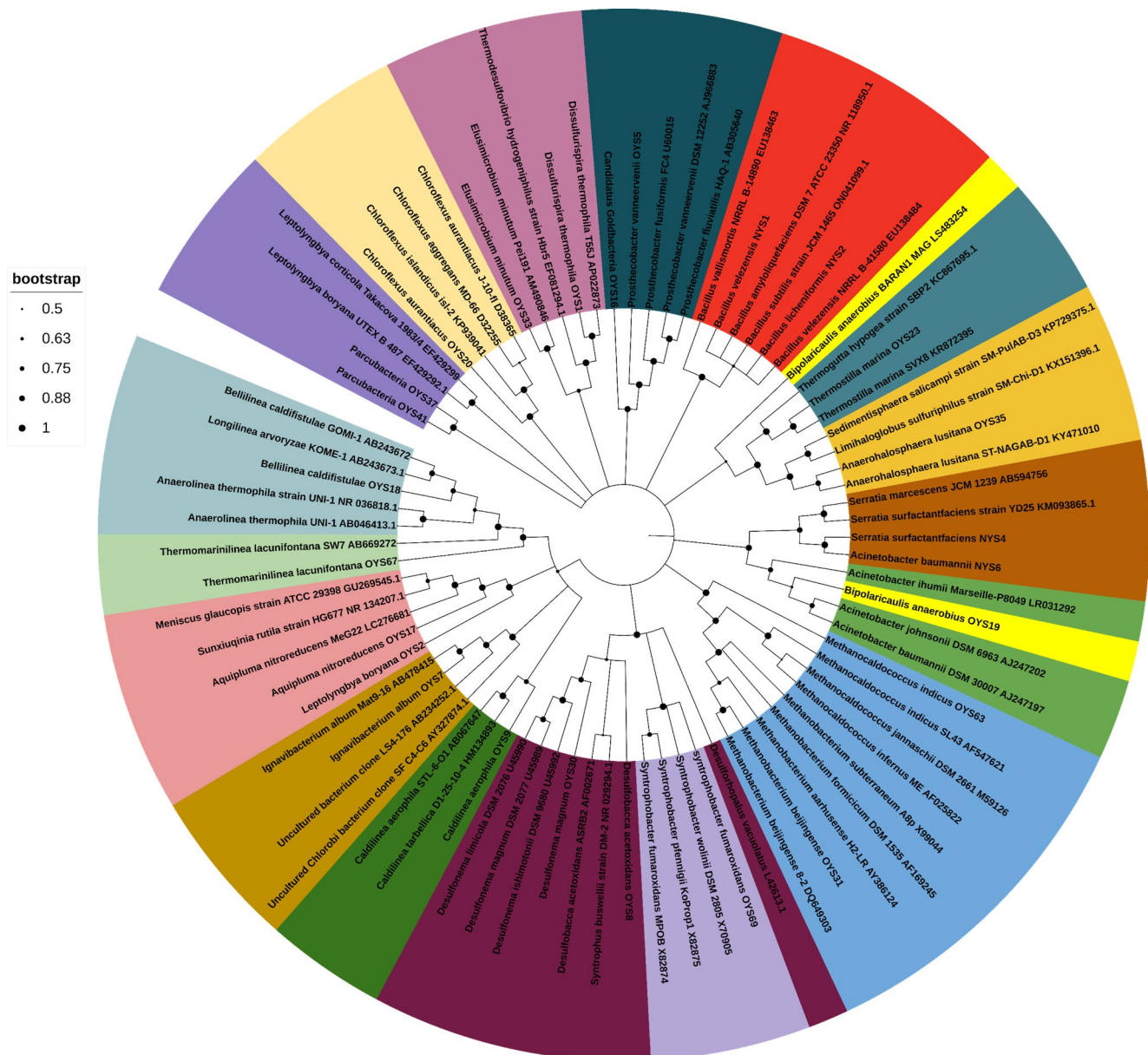
Sample name	NYS	OYS
Assembled Genome Size (bp)	67,950,615 (~ 67 Mbp)	428,411,900 (~ 428 Mbp)
Total sequences/contigs	38,986	456,587
Minimum sequence length (bp)	56	44
Maximum sequence length (bp)	2,031,872	1,302,649
N50 length (bp)	422,204	3202
G + C% observed in assembled genome	47.73%	55.23%

Table 3 Taxonomic classification and identification of various MAGs

Isolates	Domain	Phylum	Class	Order	Family	Genus	Species
OYS1	Bacteria	Nitrospirae	Thermodesulfobionia	Thermodesulfobionales	Dissulfurispiraceae	Dissulfurispira	<i>Dissulfurispira thermophila</i>
OYS2	Bacteria	Cyanobacteria	Synechococcales	Leptolyngbyales	Leptolyngbyaceae	Leptolyngbya	<i>Leptolyngbya boryana</i>
OYS5	Bacteria	Verrucomicrobia	Verrucomicrobiae	Verrucomicrobiales	Verrucomicrobiaceae	Prostheco bacter	<i>Prostheco bacter vanneervanii</i>
OYS7	Bacteria	Bacteroidetes	Ignavibacteriae	Ignavibacteriales	Ignavibacteriaceae	Ignavibacterium	<i>Ignavibacterium album</i>
OYS8	Bacteria	Proteobacteria	Deltaproteobacteria	Syntrophales	Syntrophaceae	Desulfobacca	<i>Desulfobacca acetoxidans</i>
OYS9	Bacteria	Chloroflexi	Caldilineae	Caldilineales	Caldilineaceae	Caldilinea	<i>Caldilinea aerophila</i>
OYS16	Bacteria	Candidatus Goldbacteria	—	—	—	—	—
OYS17	Bacteria	Bacteroidetes	Bacteroidia	Bacteroidales	Prolixibacteraceae	Aquipluma	<i>Aquipluma nitroreducens</i>
OYS18	Bacteria	Chloroflexi	Anaerolineae	Anaerolineales	Anaerolinaceae	Bellilinea	<i>Bellilinea caldifistulae</i>
OYS19	Bacteria	Bipolaricaulota	Bipolaricaulia	Bipolaricaulales	Bipolaricaulaceae	Bipolaricaulis	<i>Bipolaricaulis anaerobius</i>
OYS20	Bacteria	Chloroflexi	Chloroflexia	Chloroflexales	Chloroflexaceae	Chloroflexus	<i>Chloroflexus aurantiacus</i>
OYS23	Bacteria	Planctomycetes	Planctomycetia	Pirellulales	Thermoguttaceae	Thermostilla	<i>Thermostilla marina</i>
OYS30	Bacteria	Proteobacteria	Deltaproteobacteria	Desulfobacteriales	Desulfobacteraceae	Desulfonema	<i>Desulfonema magnum</i>
OYS33	Bacteria	Elusimicrobiota	Elusimicrobia	Elusimicrobiales	Elusimicrobiaceae	Elusimicrobium	<i>Elusimicrobium minutum</i>
OYS35	Bacteria	Planctomycetes	Phycisphaerae	Sedimentisphaerales	Anaerohalophaeraceae	Anaerohalosphaera	<i>Anaerohalosphaera lusitana</i>
OYS37	Bacteria	Parcubacteria	Paceibacter	-	-	-	-
OYS31	Archaea	Euryarchaeota	Methanobacteria	Methanobacteriales	Methanobacteriaceae	Methanobacterium	<i>Methanobacterium beijingense</i>
OYS41	Bacteria	Parcubacteria	Paceibacter	-	-	-	-
OYS63	Archaea	Euryarchaeota	Methanococci	Methanococcales	Methanocaldococcaceae	Methanocaldococcus	<i>Methanocaldococcus indicus</i>
OYS67	Bacteria	Chloroflexi	Anaerolineae	Anaerolineales	Anaerolinaceae	Thermomarinilinea	<i>Thermomarinilinea lacunofontalis</i>
OYS69	Bacteria	Proteobacteria	Deltaproteobacteria	Syntrophobacteriales	syntrophobacteraceae	Syntrophobacter	<i>syntrophobacter fumaroxidans</i>
NYS1	Bacteria	Firmicutes	Bacilli	Bacillales	Bacillaceae	Bacillus	<i>Bacillus velezensis</i>

Table 3 (continued)

Isolates	Domain	Phylum	Class	Order	Family	Genus	Species
NYS2	Bacteria	Firmicutes	Bacilli	Bacillales	Bacillaceae	Bacillus	<i>Bacillus licheniformis</i>
NYS4	Bacteria	Proteobacteria	Gammaproteobacteria	Enterobacterales	Enterobacteriaceae	Serratia	<i>Serratia surfactantifaciens</i>
NYS6	Bacteria	Proteobacteria	Gammaproteobacteria	Pseudomonadales	Moraxellaceae	Acinetobacter	<i>Acinetobacter baumannii</i>

**Fig. 2** Maximum likelihood phylogenetic tree was constructed using MEGA 11 software with bootstrap values of 500. The phylogenetic tree was the edited and modified in iTOL interactive tree of life.

Carbohydrate active enzyme (CAZyme) gene prediction

The filtered final binned genome, were subjected to CAZyme gene prediction using dbCAN meta server. The results have shown the presence of various carbohydrate active enzymes such as Carbohydrate Esterase family (CE), Glycoside Hydrolase (GH), Glycosyl Transferase (GT) and Polysaccharide Lyase (PL). It was found that the major enzyme present in almost all the MAGs was Glycosyl Transferase with average value of (56.7%) followed by Glycoside Hydrolase (37.4%), Carbohydrate Esterase family (8.2%) and Polysaccharide Lyase (1.9%). The distribution of genes encoding carbohydrate degrading enzymes in the individual genomes are summarized in Supplementary Tables 4, Fig. 3. On average, each member of the phylum Chloroflexi encodes the most carbohydrate degrading enzymes (324) followed by Planctomycetes (170), Cyanobacteria (137) and Proteobacteria (107). Furthermore, the binned genome was analyzed through RefSeq using Kbase tool to extract various CAZymes. The four types of carbohydrate active enzymes such as Carbohydrate Esterase (CE), Glycoside Hydrolase (GH), Glycosyl Transferase (GT) and Polysaccharide Lyase (PL) were checked in the MAGs. The results reveal that the binned genomes contained majority of GT-Glycosyl transferase; followed by GH-Glycoside Hydrolase; PL-Polysaccharide Lyase and CE-Carbohydrate esterase were observed

in very low number. NCBI BlastX results reveal that the majority of CAZymes found belonged to different bacterial species, however, the similarity indices against the protein were found to be significantly low (Supplementary Excel file 1).

Antibiotic resistance gene profiling

Antibiotic resistance gene profiling was done using CARD database. Very less antibiotic resistance genes were found. Out of three NYS binned genome, one genome had antibiotic resistance gene in them. NYS4 (*Serratia surfactantfaciens*) was found to possess different resistance gene classes such as Aminoglycosides, Carbapenem, Cephalosporin, Fluoroquinolones, and Tetracycline. However, out of 21 OYS binned genome four were harboring antibiotic resistance gene in them. Only two antibiotic classes Fluoroquinolones, Tetracycline were found to be present in all the four OYS genomes such as OYS2 (*Leptolyngbya boryana*), OYS5 (*Prosthecobacter vanneervanii*), OYS30 (*Desulfonema magnum*), OYS37 (Unknown) (Table 4).

Heavy metal gene screening

The predicted gene protein sequence for all the binned genome of NYS and OYS sample were aligned against the BacMet experimentally confirmed database using

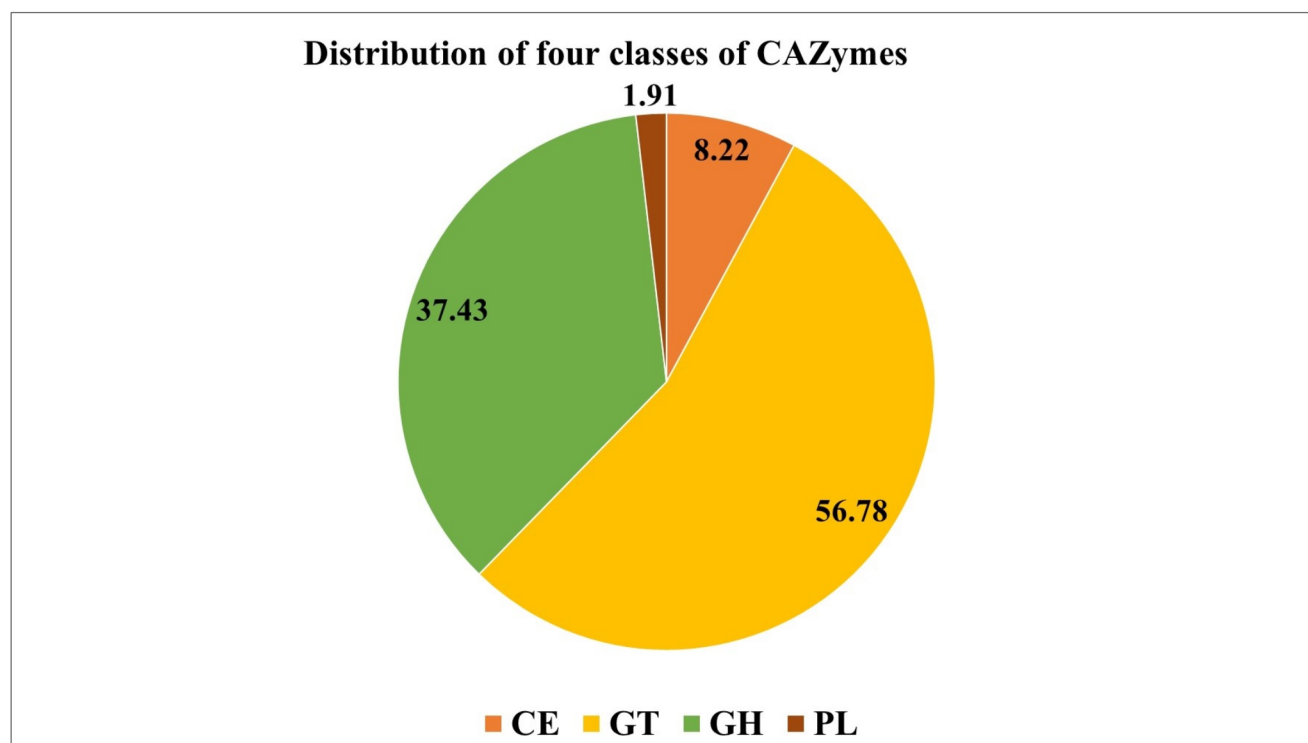


Fig. 3 The percent distribution of four classes of CAZymes. CE-Carbohydrate esterases; GT-Glycosyl transferases; GH-Glycoside hydrolases; PL-Polysaccharide lyases.

Table 4 Antibiotic Resistance gene Profiling Count for NYS and OYS samples

Resistance Antibiotic Name	Total number of ARGs in NYS-004	Total number of ARGs in OYS-002	Total number of ARGs in OYS-005	Total number of ARGs in OYS-030	Total number of ARGs in OYS-037
Aminoglycosides	1	n.f.	n.f.	n.f.	n.f.
Carbapenem	1	n.f.	n.f.	n.f.	n.f.
Cephalosporin	2	n.f.	n.f.	n.f.	n.f.
Cepharmycin	1	n.f.	n.f.	n.f.	n.f.
Elfamycin	1	n.f.	n.f.	n.f.	n.f.
Fluoroquinolones	4	1	1	1	1
Macrolides	1	n.f.	n.f.	n.f.	n.f.
Monobactam	1	n.f.	n.f.	n.f.	n.f.
Penam	2	n.f.	n.f.	n.f.	n.f.
Tetracycline	3	1	1	1	1

*where n.f. refers to not detected/not found

Table 5 Heavy metal genes found in the binned NYS genome

Gene name	NYS-002	NYS-004
<i>arsR</i>	2	n.f.
<i>kpmO</i>	n.f.	2
<i>mdtG/yceE</i>	n.f.	2
<i>pstB</i>	n.f.	2
<i>robA</i>	2	n.f.

*where n.f. refers to not detected/not found

Table 6 Heavy metal genes found in the binned OYS genome

Bins	Gene Name				
	<i>arsB</i>	<i>ruvB</i>	<i>pstB</i>	<i>sodA</i>	<i>dpsA</i>
OYS-001	n.f.	2	n.f.	n.f.	n.f.
OYS-002	1	n.f.	1	n.f.	1
OYS-004	n.f.	1	2	n.f.	1
OYS-005	n.f.	n.f.	n.f.	1	n.f.
OYS-007	1	n.f.	1	n.f.	n.f.
OYS-008	1	n.f.	n.f.	n.f.	n.f.
OYS-009	1	n.f.	1	1	n.f.
OYS-012	1	n.f.	n.f.	n.f.	n.f.
OYS-016	1	n.f.	n.f.	n.f.	n.f.
OYS-017	1	n.f.	n.f.	n.f.	n.f.
OYS-020	1	1	n.f.	n.f.	n.f.
OYS-023	1	n.f.	n.f.	1	n.f.
OYS-031	n.f.	2	n.f.	n.f.	n.f.
OYS-033	1	n.f.	n.f.	n.f.	n.f.

*where n.f. refers to not detected/not found

BLASTp algorithm. A total of 110 heavy metal genes (of which 102 unique genes) was found in two NYS binned genome; 4 for NYS2 (*Rickettsiella*) and 106 for NYS4 (*Serratia surfactantifaciens*) respectively. Zinc (Zn); Arsenic (As) and Copper (Cu) were the top three heavy metals for which the gene has been found in the binned genome (Table 5a). For OYS sample, a total of 48 heavy metal genes (of which 24 was with unique genes) was found in 14 binned genomes (OYS1; OYS2; OYS4; OYS5; OYS7; OYS8; OYS9; OYS12; OYS16; OYS17; OYS20; OYS23; OYS31; OYS-33). Arsenic (As); Chromium (Cr); Tellurium (Te); Selenium (Se); Benzylkonium Chloride (BAC) [class:

Quarternary Ammonium Compounds (QACs)] were the top-heavy metals for which the gene has been found in the binned genomes (Table 6).

Sulfur metabolism

It has been shown that the studied genomes possess various genes related to sulfur metabolism. Almost all the genomes possess Thioredoxin reductase (EC 1.8.1.9) (Trxr gene) found with the genomic annotation using SEED RAST. Various other genes for enzymes such as Thiol peroxidase, Tpx-type (EC 1.11.1.15), Thiol peroxidase, Bcp-type (EC 1.11.1.15) and Alkyl hydroperoxide reductase subunit C-like protein (AhpC) were found in many genomes of both the OYS and NYS hot springs. Alkanesulfonates assimilation genes such as *ssuA*, *ssuC*, *ssuD*, *ssuE*, were also present mostly in genomes of NYS hot spring. Inorganic Sulfur Assimilation was also found within the genomes. Various enzymes or proteins present were Sulfate and thio-sulfate import ATP-binding protein CysA (EC 3.6.3.25) (CysAWTP), Sulfate-binding protein Sbp, Sulfate transporter, CysZ-type, Sulfate adenylyltransferase subunit 1 (EC 2.7.7.4) (SAT), Sulfite reductase [NADPH] flavoprotein alpha-component (EC 1.8.1.2) (SIR_FP), Sulfite reductase [NADPH] hemoprotein beta-component (EC 1.8.1.2) (SIR_HP). Sulfite reduction-associated complex DsrMK-JOP and co-clustering genes such as DsrA, DsrB, DsrM, DsrK, DsrJ and DsrO were also present in various genomes. The clustering of sulfur metabolizing enzymes or proteins and their representation is shown by heat map Fig. 4.

Nitrogen metabolism

Similar to sulfur metabolism, various genes were found to represent nitrogen metabolism. Nitrosative stress protection systems were found to be present such as Nitrite-sensitive transcriptional repressor NsrR. Nitrate and nitrite ammonification related genes were also found to be present such as Nitrite reductase [NADPH] large subunit (EC 1.7.1.4)

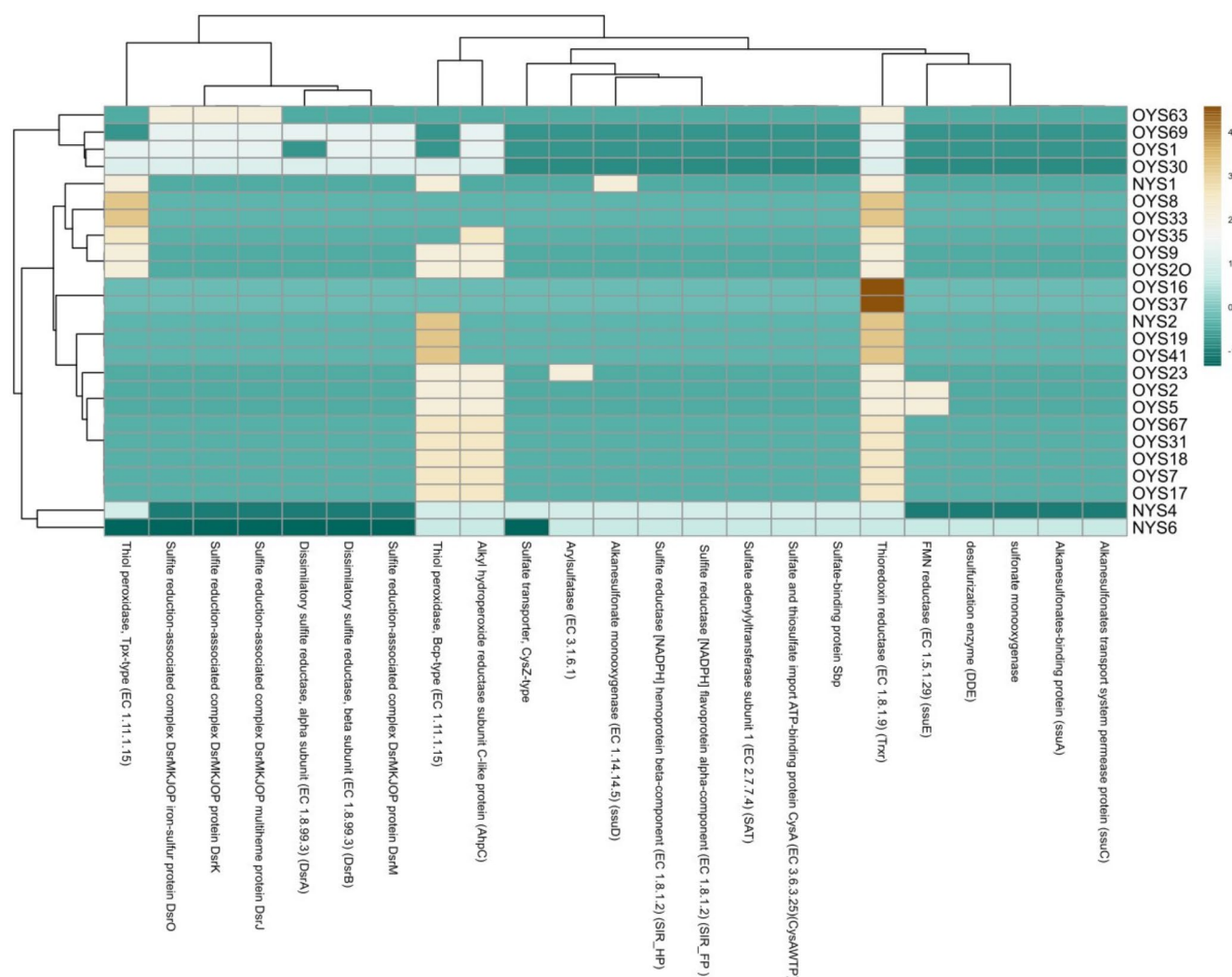


Fig. 4 Heat map, showing the distribution of sulfur metabolism related genes.

(NiRas1), Nitrite reductase [NADPH] small subunit (EC 1.7.1.4) (NiRas2), Nitrate/nitrite transporter (NiT), Nitrate/nitrite transporter (NaNiT), Respiratory nitrate reductase gamma chain (EC 1.7.99.4) (NaRres) and Nitrate/nitrite sensor protein (EC 2.7.3.-) (Reg). Various ammonia assimilation related enzymes were also found in various studied genomes. Glutamine synthetase type I (EC 6.3.1.2) (GS), Glutamate-ammonia-ligase adenylyltransferase (EC 2.7.7.42) (GlnE), Glutamate synthase [NADPH] small chain (EC 1.4.1.13) (NADPH_GOGAT), Ferredoxin-dependent glutamate synthase (EC 1.4.7.1) (GOGATF), Nitrogen regulatory protein P-II and Ammonium transporter (amt) were found to be present. Denitrifying reductase gene clusters such as NarK, NarG, NarH, NarJ, NorD, NorQ were also present in many studied genomes of both NYS and OYS hot springs. The clustering of nitrogen metabolizing enzymes or proteins and their representation is shown by heat map Fig. 5.

Discussion

Many lineages identified from 16S rRNA sequencing surveys still lack genomic representation despite significant advances in the field of taxonomy (Parks et al. 2017). However, a lot of the information from pure culture genome sequencing can be accelerated via metagenomics (Hugerth et al. 2015). Researchers now have the unheard-of opportunity to collect sequences from thousands of genomes at once in natural samples with the advent to the revolution in throughput DNA sequencing and their lower costs that has occurred over the past ten years. Despite the enormous amount of sequencing data that may be used to infer information on the distribution of biological lineages and metabolic potentials worldwide (Dupont et al. 2014; Venter et al. 2004), many biogeochemical, ecological, and evolutionary analyses call for the organizing of data into genomes. From naturally occurring microorganisms, genomes can serve as

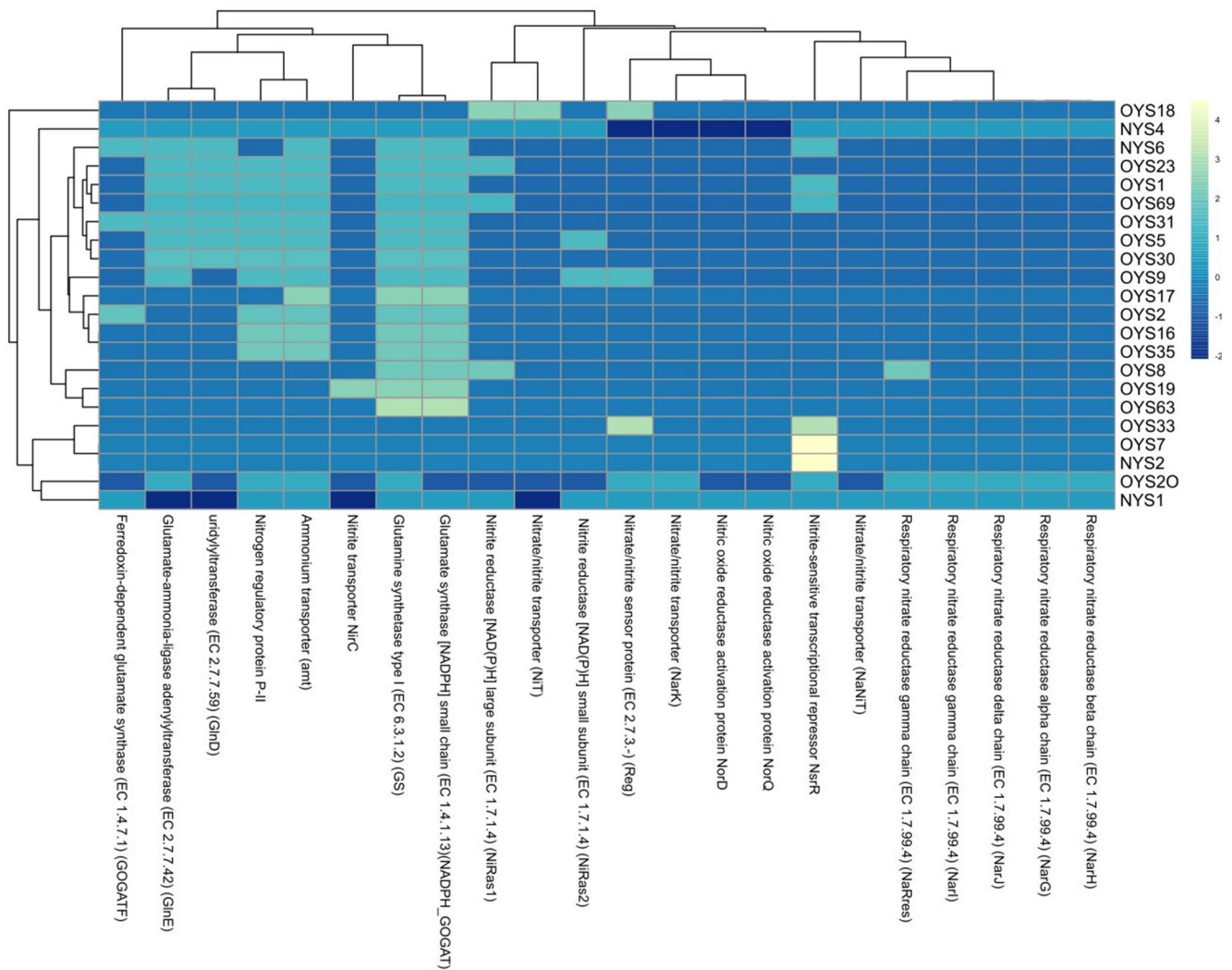


Fig. 5 Heat map, showing the distribution of nitrogen metabolism related genes.

references, enabling high-quality annotations to be produced in ensuing high-throughput environmental investigations where, in the absence of such references, the majority of sequences or peptides would remain unidentified (Hug-erth et al. 2015; Ottesen et al. 2013). Contigs or scaffolds from the same lineage can be grouped and genomes can be rebuilt through the process of binning. Numerous ecosystems, including aquatic ones such as hot springs, have effectively used the metagenome binning method (Abe et al. 2005; Baker et al. 2015; Podell et al. 2013). Without the aid of reference genomes, first approaches based on sequence composition (such as tetranucleotide frequencies) were effective in reconstructing almost whole genomes from metagenomic contigs (Abe et al. 2005; Dick et al. 2009). However, these techniques often can only distinguish between taxa. Binning down to species and occasionally strain level has been made possible more recently by

the utilization of coverage variation across many samples (Albertsen et al. 2013; Nielsen et al. 2014).

Geothermal features such as hot springs have been studied throughout the world using both culture dependent and culture in-dependent techniques (Najar et al. 2022b). However, there are very few studies done on the construction of MAGs via high throughput sequencing from water or soil samples of hot springs. The present study is the only and recent studies on the construction of MAGs via metagenomics among the hot springs present in Himalayan Geothermal Belt. Hot springs are the hot spots and represent the repositories of huge range and diversity of microbes (Najar et al. 2018). The microbes present in hot springs are usually thermophiles or hyperthermophiles. Thermophiles are the microbes that can with stand the higher temperatures often more than 45 °C, with optimum growth at 50–60 °C (Bertoldo and Antranikian 2002; Panda et al. 2018). Various metagenomic studies have shown the predominance of

phylum such as Proteobacteria, Firmicutes, Bacteroidetes, Chloroflexi etc. in hot springs throughout the world (Najar et al. 2018). In the present study, MAGs were constructed and it was found that the predominant phyla were Proteobacteria, Chloroflexi and Firmicutes, thus similar to the metagenomic amplicon sequencing.

The two hot springs i.e., OYS and NYS soil samples considered for the study were explored to construct their MAGs from the metagenomic sequencing data. The sequencing data was curated and the metagenomic bins with 80% completeness and 20% contamination was considered as putative individual genome and was screened for the presence of 16S rRNA gene for binned genome. Apart from the dominance of the phylum Proteobacteria, Chloroflexi and Bacteroidetes there were other geothermal soil intrinsic bacteriomes belonging to Nitrospirae, Cyanobacteria, Verucomicrobia, Bipolaricaulota, Planctomycetes, Elusimicrobiota, and Parcubacteria. There was also Euryarchaeota representatives which is a very common phylum in the geothermal regions. The 25 MAGs found here in OYS and NYS hot spring sediments have been also reported from various worldwide geothermal regions or natural extremophilic areas. These bacteria are regarded as the signature genomes that are generally present in sediments.

The phylum Proteobacteria currently represents the largest and most phenotypically diversified evolutionary branch within the domain Bacteria. In order to highlight the tremendous diversity of morphologies and physiologies seen within this bacterial phylum, Stackebrandt et al. in 1988 named the Proteobacteria after the Greek god Proteus, who could take on many various shapes (Gupta 2000). The majority of the known Gram negatives of interest in medicine, veterinary medicine, industry, and agriculture belong to the Proteobacteria which make up more than 40% of all validly published prokaryotic genera. They also represent a significant portion of the traditional Gram-negative bacteria, exhibit extreme metabolic diversity, and have significant biological significance (Kerstens et al. 2006). Phylum Proteobacteria MAGs included *Desulfobacca acetoxidans*, *Desulfonema magnum*, *Syntrophobacter fumaroxidans*, *Serratia surfactantfaciens*, and *Acinetobacter baumannii*. *Desulfobacca acetoxidans* *Desulfonema magnum* *Syntrophobacter fumaroxidans* *Serratia surfactantfaciens* *Acinetobacter baumannii*.

Phylum Chloroflexi had the representatives from *Chloroflexus aurantiacus*, *Caldilinea aerophila*, *Bellilinea caldifistulae*, and *Thermomarinilinea lacunofontalis*. The Chloroflexi is a broad phylum of bacteria with a variety of metabolic characteristics. Chloroflexi, Anaerolineae, Caldilineae, Ktedonobacteria, Dehalococcoidetdia, and Thermomicrobia are the six classes that make up this phylum (Hanada 2014). The majority of classes have organisms with filamentous morphology as a defining characteristic

feature. Only the class Chloroflexi, out of these six, contains phototrophic bacteria. Typically, alkaline to neutral hot springs are habitat to *Chloroflexus* species. *Chloroflexus aurantiacus* (Pierson and Castenholz 1974) and *Cfl. aggregans* (Hanada et al. 1995) have both been isolated from hot springs and are the only two valid species in this genus. The organisms create yellow-orange greenish mats with cyanobacteria in natural hot springs. Depending on the temperature, pH, and sulfide concentration, the mats' form and color can change (Hanada 2014). Thin orange films or yellow-orange streamers that have grown on the surface of dark green cyanobacterial mats are the *Chloroflexus* mats, which can be seen in hot spring effluents. Thus, all the observed MAGs related to OYS sediments are the natural habitats for these species.

Planctomycetes have been reported from a variety of environmental habitats, such as soil, hot springs, salt pits, and peat lands, marine and freshwater sediments, and various anoxic regions like wastelands, and coastal sediments. (Fuerst and Sagulenko 2011; Slobodkina et al. 2016). *Thermotilla marina* is an anaerobic planctomycete that was first isolated from the hydrothermal vent of Vulcano Island of Italy (Slobodkina et al. 2016). The optimal conditions required for its growth was reported to be 55 °C, in moderate brine alkaline conditions (pH 8–9). Thus, similarly here in the OYS sediments its MAG was found as it had the most suitable niche for its growth. Another planctomycetes found in OYS was *Anaerohalosphaera lusitana*. This planctomycete is an exclusive resident of anoxic sediments. It was first reported from the solar salterns of Tavira, Portugal (Pradel et al. 2020). This bacterium requires mesophilic to thermophilic temperature of growth with high saline and alkaline conditions.

The lesser abundant MAGs of the various understudied bacterial phyla representatives were the species related to *Dissulfurispira thermophila*, *Leptolyngbya boryana*, *Prostheco bacter vanneervanii*, *Ignavibacterium album*, *Aquipluma nitroreducens*, *Bipolaricaulis anaerobius*, and *Elusimicrobium minutum*. *Dissulfurispira thermophila* belongs to phylum Nitrospirae. This is an obligate anaerobic thermophile that grows by reducing sulfate compounds to sulfide and requires organic carbon for metabolism (Umezawa et al. 2021, 2022). *Leptolyngbya boryana* is the only Cyanobacteria MAG found in OYS sediments. Cyanobacteria are the oxygen-evolving photosynthetic bacteria that can be found on the surface of freshwater, ocean, brackish water, and soil. They can also be found in harsh conditions like hot springs and polar-regions (Keshari et al. 2022; Mukherjee et al. 2022). The genus *Leptolyngbya* has been isolated from a variety of habitats, including soil surfaces. These thin filamentous cyanobacteria are distinguished by the low width of their cylindrical trichomes (Shimura et al.

2015). Their most importance significance as a primary producer is not limited to the environment only. It has been recently been reported to show good biosorption potential (Yadav et al. 2021). *Prostheco bacter vanneervanii* belongs to Verrucomicrobia phylum. It is a heterotrophic plankton usually found in oligotrophic habitats (Hedlund et al. 1997). Several other has been reported from activated sludges (Lee et al. 2014) and Himalayan freshwater lakes in Kashmir. *Ignavibacterium album* and *Aquipluma nitroreducens* belongs to phylum Bacteroidetes. *Ignavibacterium album* is a chemoheterotrophic sulfur reducing thermophilic bacteria first discovered from the solfataric hot spring thermal mats (Iino et al. 2010). *Aquipluma nitroreducens* is nitrate-reducing, facultatively anaerobic bacteria discovered from a freshwater Japanese lake (Watanabe et al. 2020). *Bipolaricaulis anaerobius* is a prosthecae bacterium discovered from anaerobic digesters (Hao et al. 2018) belonging to phylum Bipolaricaulota. *Elusimicrobium minutum* belonging to phylum Elusimicrobiota is the most interesting observation in the MAGs of OYS. It has been reported as termite and mammalian gut bacterium (Geissinger et al. 2009; Herlemann et al. 2009) and most probably its genome got contaminated with the investigating solfataric soil. It is yet to be established as an ecological flora of any geothermal systems. A complex phototrophic microbial mat sample taken from a spring in the Great Boiling Spring (GBS) Geothermal Field in Gerlach, Nevada, was used to discover a cellulose-degrading microbial consortium comprising of Candidatus Goldbacteria (Doud et al. 2020). Detection of class Paceibacteria MAGs in OYS was another interesting finding as it has been reported from various freshwater lakes and deep groundwater (Chaudhari et al. 2021; Chiriac et al. 2022). The detection of these MAGs shows very overwhelming prospects of future discoveries of novel microbial species. Their detection might be attributed to the factor that they are natural ecological residents of geothermally heated regions or freshwater lakes or natural soil flora having 40–60 °C as suitable temperature for their growth accompanied by alkaline rich sediments (pH 7.8–9) and rich organic sources for metabolism.

The Euryarchaeota MAGs comprised of *Methanobacterium beijingense* *Methanocaldococcus indicus* in OYS sediments. These are methanogens. They live in a wide variety of settings, including freshwater and marine sediments, animal gastrointestinal tracts, and anaerobic waste digesters, despite having a very identical energy metabolism (Das et al. 2020b). Methane production as the sole byproduct of energy metabolism and strictly anaerobic growth are the two main physiological traits that most of the methanogens show (Hanišáková et al. 2022). *Methanobacterium beijingense* was first reported from Beijing, China (Ma et al. 2005). This methanogen can survive temperatures of 50 °C

and alkaline conditions (pH 8–9) similar to the OYS geography. Likewise, *Methanocaldococcus indicus* is a hyperthermophile autotrophic methanogen that was first discovered from a deep-sea hydrothermal chimney located at Central Indian Ridge, Indian Ocean (L'Haridon et al. 2003). It optimally grows in 50–80 °C alkaline and sulfurous conditions (pH 8–9). Thus, it is interesting to note that the highest recorded temperature of OYS hot spring was 72.3 °C and with its alkaline and sulfur rich conditions, it seems to be very good natural incubator for such kind of methanogens. Presence of methanogenic MAGs suggests that these geothermal niches should be explored further to isolate such industrially important methanogens.

One of the studies was done on two thermal pools in Kamchatka, Russia by Wilkins and his research team (Wilkins et al. 2019). They have constructed MAGs and have found the abundance of phylum Aquificae, Proteobacteria, Firmicutes and Bacteroidetes. They have also found some archaea such as Crenarchaeota and Thaumarchaeota. The similar study was carried out on brackish water by Hugerth et al. 2015. They have found that the reconstructed genomes belong to Actinobacteria, Bacteroidetes, Cyanobacteria, Verrucomicrobia, Alpha-, Beta- and Gamma-proteobacteria and Thaumarchaeota (Hugerth et al. 2015).

Physicochemical analysis shown in the present study (Table 1) and other study (not yet published) showed that there is no considerable distinction between these two considered hot springs (Das et al. 2020a, b, c, 2021). Thus, it may be predicted that the diversity found through construction of MAGs is usually temperature dependent. Since temperature is the primary defining factor of all hot springs, several researchers have connected and interpreted the abundance of a certain phylum in the hot springs as a function of temperature. For example, in the hot springs of Odisha, India, Sahoo et al. (2017) connected and linked the major Proteobacteria species as a function of temperature (Sahoo et al. 2017). In Indian hot springs, where the temperature ranges from 50 to 70 °C, Firmicutes prevail. Additionally, it was discovered that the phylum Firmicutes is the most adaptable in hot springs and may be able to thrive in a variety of temperature gradients (Najar et al. 2022a). However, the present study has found mixture of thermophilic and mesophilic bacteria. This might be the due to the contamination of upper soil layers of hot springs (Najar et al. 2022a).

Antibiotic resistance was also checked in the MAGs corresponding to OYS and NYS. It was found that there is very less or considerably no antibiotic resistance. One of the MAG NYS4 has found to possess antibiotic resistance. NYS4 was identified as *Serratia surfactantfaciens* which belongs to *Enterobacteriaceae* family and is usually found in water and soil. The antibiotic resistance in such a bacterium may have gained through environment. It has been

known that this bacterium may produce a red pigment identified as prodigiosin. Prodigiosin has been shown to have antimicrobial (antifungal, antibacterial, antiprotozoal) properties (Su et al. 2016). This, suggests that the MAG NYS4 may thus possess an antibiotic resistance. However, in general the antibiotic susceptibility may also be correlated with the temperature of hot springs. Based on our earlier investigation we have correlated the antibiotic susceptibility as temperature dependent. It has been suggested that, hot springs are distinct ecosystems with less anthropogenic activity and at high temperatures, they may lack stable antibiotics. Thus, these thermophilic microbes may have less pressure or competition to acquire ARGs as a result of the absence of antibiotics (Najar et al. 2022a).

Functional metagenomics analysis was also carried to investigate the presence of genes related to sulfur and nitrogen metabolism in all the MAGs. Since the hot springs of Sikkim are rich in sulfur content (Najar et al. 2018, 2020a, b). Thus, it was reasonable to check the sulfur metabolism in these MAGs. Thioredoxin reductase (EC 1.8.1.9) was found in almost all the MAGs. It has two kinds one found in Bacteria, Archaea and most Eucarya, and one found in animals. Thioredoxin reductase involves in the conversion of Thioredoxin-disulfide in to Thioredoxin (reduced) in presence of NADH. For cells to survive, the extracellular or intracellular environment must be kept in a suitable disulfide state. The NADPH-dependent-TR/Trx system is the primary protein system that regulates the thiol/disulfide redox balance in the cytoplasm as reductants of intracellular disulfides. By lowering thioredoxin, thioredoxin reductase (TR) controls the intracellular redox environment (Trx). Recent research shows that the Trx redox network is involved in both the protection of cells from O₂ and the global redox control of metabolism in anaerobes (Valette et al. 2017). Similarly, Alkyl hydroperoxide reductase subunit C (AhpC) is the catalytic subunit in charge of detoxifying reactive oxygen species that develop in bacterial cells or come from the host (Wang et al. 2013). This enzyme was also found in most of the MAGs. Other enzyme complexes were found in many MAGs such as sulfite reduction associated complexes which are known to use sulfur compounds in a series of oxidation or reduction reactions, thereby generating metabolic energy in the process of dissimilatory metabolism (Oliveira et al. 2011). Thus, by having such sulfur metabolic mechanisms, the microbes are able to generate metabolic energy, protect cells from O₂ and help them survive under adverse conditions in these geothermal features or when they are infected.

Nitrogen is a component that all living things require. The biogeochemical nitrogen cycle is formed by the inter-conversion of nitrogen species between a range of redox states (+5 to -3) and has a variety of effects on the environment and industrial uses (Holland and Weitz 2001). Prior to

being further integrated into organic nitrogen by glutamine synthase, inorganic nitrogen oxides must first be converted to ammonium. In oxygen-limited settings, soluble nitrogen oxides, such as nitrate and nitrite, can be used by bacteria as terminal respiratory oxidants in redox reactions to generate metabolic energy. Ammonification and denitrification, two distinct nitrate respiration mechanisms, start with the same intermediate, nitrite, but produce different end products, ammonia and gaseous nitrogen oxides (NO or N₂O) or dinitrogen, respectively (Doroshchuk et al. 2006). The studied MAGs possess various enzymes related to nitrogen metabolism such as Nitrate reductase, Glutamate synthase, and Glutamine synthetase. Using hydroxylamine, ammonia-oxidizing bacteria convert ammonia to nitrite, which is then further oxidized to nitrite to nitrate by chemolithoautotrophic bacteria. This two-step process is known as autotrophic nitrification. Nitrate reductase NiRs have been found in various MAGs within the phyla Proteobacteria and Clostridia. It has been known that among soluble NiRs, most widely distributed is a siroheme containing NADPH dependent enzyme. Cyanobacteria, plants and some α -proteobacteria possess a distinct cytoplasmic NiR, which is ferredoxin-dependent. Some strictly anaerobic species (i.e. Clostridia), have another two-component NiR, which is not yet characterized (Doroshchuk et al. 2006). Thus, sulfur and nitrogen metabolic enzymes and their pathways are important in these bacteria in order to generate metabolic energy and to allow them to survive in such extreme environments.

Besides, glycoside hydrolases, polysaccharide lyases, and carbohydrate esterases are the primary enzyme activities accountable for the breakdown of complex carbohydrates among the known biomass-degrading enzymes. Thus, for industrial uses including biocatalysts, biorefining, textiles, and food processing, the identification of these degradative enzymes is extremely important (Zhou et al. 2020). The other important factor is the thermostable nature of these degradative enzymes as their thermostability can be exploited in many industrial applications. Thus, we also investigated the prevalence of various CAZymes in the considered MAGs. We found that the predominant enzymes were GT-Glycosyl transferases (56%) followed by GH-Glycoside hydrolases (37%), CE-Carbohydrate esterases (8%) and PL-Polysaccharide lyases (1.9%). Results of MAGs analysis through Kbase pipeline reveal that the occurrence of GT-Glycosyl transferases is highest in the MAGs, followed by GH-Glycoside hydrolases; PL-Polysaccharide lyases and CE-Carbohydrate esterase were found to be comparatively low. Considering a threshold $\geq 90\%$ protein sequence identity, the CAZymes found from the MAGs belonged to Cyanobacteria and Pseudomonadota. The primary contributors of the most abundant Phylum Cyanobacteria (pertaining to CAZymes according

to NCBI BlastX results) were *Leptodesmis sichuanensis*, a novel species found to colonise thermal habitats such as hot springs (Tang et al. 2022). Overall, NCBI BlastX analysis reveal that most of the CAZymes found showed low similarity (%) with the annotated proteins and most of the annotated proteins belonged to Candidatus sp. of bacteria. Thus, there remains a huge scope of identifying novel CAZymes with intricate mechanisms of thermostability. The potential novel enzymes may find their use in various biotechnological applications. Majority of biomass-degrading microbes in hot springs are yet to be discovered and with the advent of culture-independent techniques, a more detailed study pertaining to CAZyme discovery may facilitate in discovery of novel enzymes. The stringent habitual conditions of thermal environment make hot springs an ideal testbed to hunt for novel bacteria and their novel enzymes with promising use in biotechnological applications.

Thus, to unearth the unknown diversity among hot springs new methods such as high throughput sequencing and the construction of MAGs through highly specific pipe lines should be followed. At the same time the exploration of their enzymatic potential may be investigated and exploited through genetic engineering. Besides, these sophisticated techniques, various novel strategies must be framed to isolate these unknown microbes through culture dependent methods. This can add to and would be a great asset to the microbial taxonomy and simultaneously to the industry.

Conclusion

High altitude hot springs residing in the vicinity of Himalayan geothermal belt were evaluated for the first time with respect to the construction and characterization of MAGs. Specific assembly and binning parameters were used to reconstruct MAGs. We presented 25 MAGs, 21 from OYS hot spring and 4 from NYS hot spring based on the criteria that 16 S rRNA gene was successfully predicted. Various databases such as GTDB, Kaiju, EzTaxon and NCBI BLAST non-redundant protein database *nr* were used to find the nearest references for the binned genomes. In the present study, we found the prevalence of various phyla such as Proteobacteria, Chloroflexi, and Firmicutes which are in accordance with the results found through high throughput amplicon sequencing from various hot springs throughout the world. We also correlated and found that the present microbial diversity and their antibiotic resistance profile may be temperature dependent. Besides, the bioactive potential of the considered MAGs was evaluated. It was found that the MAGs were rich in CAZymes and possess the genes for sulfur and nitrogen metabolism. This study finds the significant industrial potential of these MAGs and

can be enhanced through metabolic engineering. The thermostable properties of the bioactive compounds can also be exploited in these MAGs. However, significant study needs to explore, characterize and exploit the industrial potential of the microbial diversity residing in these hot springs. Also, novel strategies should be bent to isolate unknown microbial diversity from such extreme hot spots.

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Author contributions Nagendra Thakur conceived the idea and planned the study. Sayak Das and Ishfaq Nabi Najar collected the data, devised the initial draft, reviewed the final draft, and contributed equally to this study as the first author. Mingma Thundu Sherpa, Santosh Kumar, Prayatna Sharma, Krishnendu Mondal and Sonia Tamang reviewed the draft for submission. All the authors read and approved the final version of the manuscript.

Declarations

Competing interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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